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SUBJECT CODE NO: - RNP-8019

FACULTY OF SCIENCE AND TECHNOLOGY

F.E. (Group B) (NEP) Sem - I

Examination November / December 2025 (To be held in February-2026)

Basics of Electrical Engineering

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION A

Q1 Solve any 10 from the following questions. **20**

1. The energy used in one hour at the rate of 1kW is known as ______
   A. 1 Unit B. Infinite C. 1 joule D. high

2. The Temperature coefficient of Alloy is ______
   A. Positive B. Negative C. zero D. None

3. The property of magnetic circuit which opposed to flow of flux is ______
   A. Reluctance B. Resistance C. Reactance D. conductance

4. Line current in delta connected three phase ______ times line current.
   A. 3 time B. $\sqrt{3}$ C. Equal D. 2 Times

5. The value of form factor of sinusoidal waveform is
   A. 1.11 B. 1.21 C. 0.70 D. 0.63

6. RMS value of sinusoidal wave form is ______ Maximum value.
   A. 1.11 B. 1.21 C. 0.70 D. 0.63

7. The conductor and flux having angle between them 90 then flux link
   A. Maximum B. Minimum C. Equal D. Zero

8. The ratio of maximum value to the r.m.s. value of an alternating quantity is known ______
   A. Zero B. Peak Factor C. Form Factor D. None

9. The flux variations being brought about by variations
   A. Density B. voltage C. current D. Power

10. An ideal voltage source is that which has an internal resistance
    A. Zero B. Low C. High D. Infinite

11. The phenomenon of lagging of flux density (B) behind the magnetising force (H) in a magnetic material
    A. Flux B. Current C. B-H Curve D. Density

12. In Kirchhoff's first law $\sum i = 0$ at the junction is based on the conservation of
    A. Charge B. Current C. Energy D. Voltage

13. The resistance of a material is most commonly determined by four factors- length, cross-sectional area, type of material and ______
    A. Current B. Voltage C. Power D. Temperature

SECTION B 40

Q2 a) Prove the Series and parallel combination of resistor with circuit diagram. **05**
b) Expression for value of α at different temperature in detail. **05**

Q3 a) Explain temperature coefficient of according to material. **05**
b) Explain Ideal & Practical sources in detail. **05**

Q4 a) Comparison between Magnetic and Electric Circuits. **05**
b) A solenoid have 600 turns supply with 30 volt of resistance is 6 ohm, length of solenoid is 16 cm with area is 6 sq.cm, relative permeability is 800 find 1) MMF 2) Magnetic field intensity 3) Flux density 4) Reluctance **05**

Q5 a) Prove the Coefficient of Coupling with equation. **05**
b) Prove the Capacitors In Series with detail. **05**

Q6 a) Explain R-L Series A.C. Circuit with phasor diagram **05**
b) Explain R-C Series A.C. Circuit. **05**

Q7 a) Explain the relation between Star connected phase voltage and line voltage in detail **05**
b) Explain the relation between Delta connected phase current and line current in detail **05**